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Techno-juggling

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Abstract

Techno-juggling is both an interactive installation and a term to describe a community of technologists with a shared interest in juggling. In contributing to said community, I gather design ingredients through examination of juggling-automatons, memes produced with the siteswap animator Juggling Lab as shared on Tumblr, historically significant productions of Juggling in the 1980's emerging computer graphics industry, and existing approaches by Techno-jugglers to design digital props. An iterative design process follows culminating in the use of a TV as an angled roll juggling surface equipped with color sensitive ball tracking. This design challenges the dominance of VR, cultivating alternative forms of multimodal interaction with quasi-virtual objects.

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Preface

This research explores the work of "Techno-jugglers", a term describing a community of technologists with a shared passion for juggling. Claude Shannon, a digital communications pioneer and Techno-juggler, recognized a decline in Vaudeville, circuses, and juggling by the very same electronic media that his information theories helped popularize including radio, motion pictures, and television. Yet, by the 1990s, Shannon observed and participated in a revival of juggling community practice at American Universities such as the MIT Juggling Club by "algorithmically inclined"[1] peoples including mathematicians, physicists, engineers, and computer programmers. Within these juggling clubs and online communities, Techno-jugglers seek out a means to integrate their technical knowhow with the practice of juggling. As we will see, this desire manifests itself in a variety of forms such as mechanical replication through juggling machines, digital communications through siteswap animators, and even real-time audio-visual performance through camera and sensor-based ball motion tracking. This community can also be described through Christopher Kelty's term of a recursive public as "a public that is constituted by a shared concern for maintaining the means of association through which they come together as a public."[2]

The coming together of Techno-jugglers coincides with an emergence of computer graphics and home internet networks in the 1980s. Virtual scenes of simulated juggling were used by Triple-I and Pixar to demonstrate and sell their animation capabilities to the film and broadcast television industries. Subsequently, Eric Graham's Amiga Juggler animation was used by Commodore to sell their popular line of Amiga Home computers. Alongside this

[1] Shannon, "Scientific Aspects of Juggling."
[2] Kelty, *Two Bits*.

phenomenon is a popular narrative of computers as collaborators in the performance arts, and the growth of arcade and handheld home gaming systems. It is on these early Amiga and IBM home computers that Techno-jugglers first formed a recursive public through the sharing of siteswap animation programs on the Usenet newsgroup rec.juggling.

It is uncertain whether Techno-jugglers were motivated to act after their practice was marketed by said industries. Nonetheless, they integrated robotics, internet communication platforms, computer graphics into their communities in ways that maintained their means of association, that of juggling. Prior to internet video sharing, the juggling pattern language of siteswap became a means to share and invent new tricks via numerical sequences and generative stick figure animators. What is important here is that these Techno-jugglers were hobbyists, often building tools in their spare time to be shared without profits. Consequently, "they are not anticommercial or antigovernment. They exist independent of, and as a check on, constituted forms of power, which include markets and corporations"[3]. Computers equipped with siteswap software served as learning tools for jugglers and contributors in expanding the pattern lexicon and communicability of their practice; while retaining skilled body-coordination as essential to human performance.

In addition to siteswap animators, Techno-jugglers have since approached an integration of numerous other technologies within their practice including motion visuals through object tracking, musical composition through prop integrated sensors, robot automatons as juggling performers, and even to VR as an environment for simulated juggling. Not only has juggling historically held an important role in marketing an imagination for the users of emerging

[3] Kelty, *Two Bits*.

technologies, the design challenge of achieving juggling through any new medium is a key technical milestone signaling the infrastructural maturity of a platform.

Coincidentally, the PlayStation 4 platform's 2nd most popular game, "Marvel's Spider-Man"[4] produced by Insomniac Games in 2018, features a critical scene of juggling. Peter Parker works alongside his eventual villain Dr. Otto Octavius in a research lab to build a brain computer interface capable of teleoperating robotic arms. Dr. Otto Octavius's degenerative neurological disease serves as a key motivator for inventing this brain implant and tentacle armed device because it can help him retain artificial mobility in place of his diminishing motor control[5]. A key lab-test and turning point in the game's plot is when Dr. Ock makes the tentacle arms juggle near 20 tennis balls perfectly with his mind alone and no prior motor knowledge of juggling[6]. While this is science fiction depiction within a videogame, it clearly suggests the significance of juggling as a test for new technologies, especially in brain-computer interfaces for robotic teleoperation. As compiled by Luke Burrage, many engineering research groups have used juggling as a performative task for robots, machines, drones, and other automatons.[7]

Figure 1. Gameplay screenshot of Marvel's Spider-Man (Insomniac Games, 2018), played on PS4. Screenshot by author from Singularity 11's YouTube video, "Spider-Man Ps4 Doctor Octavius juggling tennis balls," April 19, 2020, https://www.youtube.com/watch?v=NRplJjBwd

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[4] *Wikipedia*, "List of best-selling PlayStation 4 video games."
[5] "Marvel's Spider-Man | Marvel's Spider-Man Wiki | Fandom."
[6] Singularity 11, *Spider-Man Ps4 Doctor Octavius Juggling Tennis Balls*.
[7] Luke Burrage, *Juggling ROBOT or NOT?*

VR gaming's headset and hand controller interactions are a relevant and timely analogue accompanying Spider-Man's imagination of teleoperation. For example, several jugglers and enthusiasts of the VR game Half-life: Alyx[8] have been able to perform a variety of common 3 and 4 ball juggling tricks through the game's portals, gravity gloves, and virtual props[9,10]. There is also a juggling-specific VR game available on the Meta Quest Platform[11]. Stephen Meschke, who has experimented with computer vision ball tracking and juggling machines, has made a VR space[12] which visualizes patterns without throwing interactions for performance[13].

Figure 2. Gameplay screenshot of VR Portal Juggling in Half-Life: Alyx (Valve, 2020), played on PS4. Screenshot by author from ChrisQuitsReality's YouTube video, "Impossible Juggling Through Portals in VR! (MIND BENDING)," April 19, 2020, https://www.youtube.com/

As a member of this Techno-juggling public, I find both interest and concern for VR forms of juggling as they question juggling's fundamental tactile relationship with physical moving props. Hands become symbolic cursors that grasp and throw virtual objects through

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[8] *Wikipedia*, "*Half-Life*."
[9] ChrisQuitsReality, *Juggling in Half-Life Alyx Is So Satisfying!*
[10] 2kliksphilip, *VR Juggling in Half-Life Alyx*.
[11] Meta, "VR Juggling on Oculus Rift."
[12] "The Juggling Edge - VR Juggling Simulator."
[13] "Juggling Data Set - Juggling Simulator."

timed handset squeezes. Body tracking technologies are central to the implementation of virtual juggling through VR platforms. This sentiment is common to VR games at large where bodily data extraction is foundational to VR platform infrastructure, and in the economic strategies of multibillion dollar VR development corporations like Meta's Reality Labs. Ben Eglison points out that:

> "In 2018, commercial systems typically track body movements 90 times per second to display the scene appropriately, and high-end systems record 18 types of movements across the head and hands. Consequently, spending 20 minutes in a VR simulation leaves just under 2 million unique recordings of body language"[14]

In a recursive public, "geeks use technology as a kind of argument, for a specific kind of order: they argue about technology, but they also argue through it"[15]. Using Soma design principles[16] I created an inclined television equipped with ball tracking to argue for alternatives to virtual or augmented reality interaction which retain the sensibilities of juggling.

Figure 3. Photo of Techno-juggling installation with pentagon ball trails, photo taken by author, March 27, 2026

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[14] Egliston and Carter, "Critical Questions for Facebook's Virtual Reality."
[15] Kelty, *Two Bits*.
[16] Höök, *Designing with the Body*.

Chapter 1: Juggler and Automaton as a Communication Pair

"Juggling seems to appeal to mathematically inclined people - Many amateur jugglers are computer programmers, majors in mathematics or physics, and the like. One knowledgeable juggler, a mathematics professor himself, told a reporter that forty per cent of jugglers are ''algorithmically inclined."
- Claude Shannon[17]

Shannon-Weaver

Lesser known about Claude Shannon is his lifelong hobby with juggling and unicycle riding. While his longstanding accomplishments in mathematics and digital communications are revered, less is acknowledged regarding his mathematical theories of juggling and construction of toy juggling machines. To further examine relationships between juggler-technologists and their juggling machines, it is useful to consider his Shannon-Weaver model of communication.

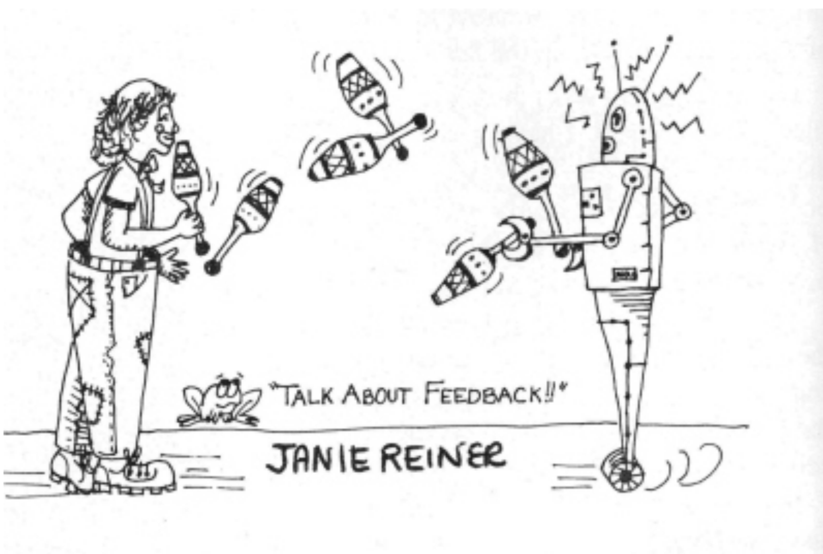

Figure 4. Talk About Feedback!!. Illustration by Janie Reiner from Juggler's World Magazine issue 34-5, December 1982. https://www.dev.juggle.org/history/archives/jugmags/images/34-5/34-5,p14-5.jpg

[17] Shannon, "Scientific Aspects of Juggling."

As applied to the illustration in Figure 1.1, the Shannon-Weaver model structures communication between a pair of senders and receivers along a channel. In digital communications, an encoding scheme is necessary to translate analog phenomena like speech into a computer readable form which may be transmitted along noisy channels and reconstructed at the receiver's output. In a Techno-juggler and automaton pair, the Techno-juggler desires to communicate analog phenomena, a 3-ball cascade, such that it can be mechanically replicated by the automaton. With the presence of feedback, a 2-way exchange is established between the pair including mathematical calculation, mechanical adjustment, and performance testing.

Reiner's depiction highlights an early imagination of robots which can participate with humans through partner club juggling. The robot is anthropomorphized with eyes, a mouth, mental processing indicated by an antenna and zig-zag lines, jointed arms, clawed hands, and wheels for feet. Claude Shannon's own realization of a juggling machine was much less humanistic than this robot imagination. The automaton's namesake to Vaudeville performer W.C. Fields reflects the comedian's legacy in inspiring "a generation of girls and boys, including the young Claude Shannon, to terrify their parents with talk of running away to join the circus."[18]

W.C Fields

Shannon's interest in juggling led him to work out mathematical equations describing the relation between dwell, flight, and vacant times based on the number of balls and hands involved. Respectively, this predicts the amount of time a ball spends in a hand, in the air, and how much time the hand is empty of a ball during one cycle of a symmetrical juggling cascade pattern. These patterns could be notated in the form of ladder diagrams which are time-based

[18] Rob Goodman, "Claude Shannon."

graphs depicting said parameters per hand over time. Shannon was able to apply these mathematical theories to design W.C Fields in the early 1980s. This was assembled by motorizing a T-shaped erector set with paddle hands to oscillate in sync with the parabolic bouncing of steel balls on a tom-tom drum. The automaton was dressed up like a ventriloquist dummy signifying the features of W.C Fields. Shannon also built a trio of motorized juggling clowns via props moving along fixed shafts[19].

Figure 5. Claude Shannon's Juggling Machines. left, W.C Fields, from MIT Museum, commons.wikimedia.org; middle, Juggling Clown Trio and right, Juggling Ladder Diagrams, from "Claude E. Shannon: Collected Papers", 1993

Inclined Plane Juggling Automatons

The mechanisms at play in W.C. Fields have remained influential for future engineers in achieving juggling through robotics systems. For instance:

> "Engineers have also built robots that juggle in two dimensions. 1980s Marc D. Donner of the IBM Thomas J. Watson Research Center used a tilted, frictionless plane, similar to

[19] Shannon, "Claude Shannon's Nodrop Juggling Diorama."

an airhockey table. It was equipped with two throwing mechanisms moving on tracks along the lower edge of the table."[20]

Contemporary Techno-jugglers Nathan Peterson and Stephen Meschke have both explored the inclined plane as space for juggling machines. Meschke's Pantograph[21] was capable of cascade juggling 5 balls, whilst Peterson's could do up to 7 as well as a 534 siteswap pattern[22].

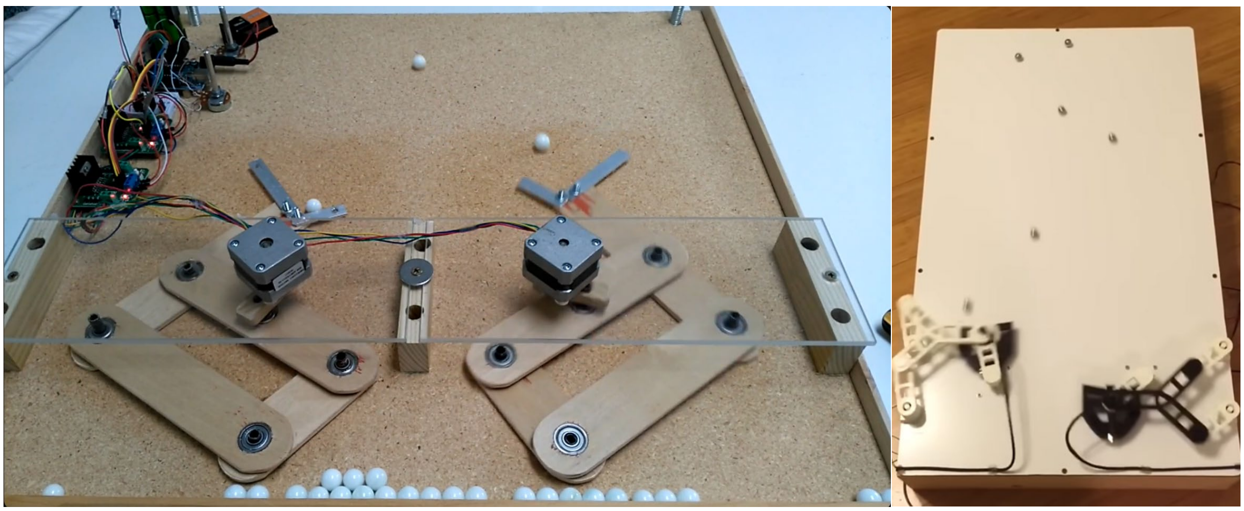

Figure 6. Angled Surface Roll Juggling Machines. Left Pantograph Juggler, Screenshot by author from Stephen Meschke's YouTube video, "Juggling Machine", July 9th, 2019. https://www.youtube.com/watch?v=uyq7RJbFZEI; Right Juggling Robot, Photo taken by creator Nathan Peterson, October 2015. https://hackaday.io/project/7727-juggling-robot

For automatons, the Shannon-Weaver model is biased towards the right. Balls are juggled in an unchanging pattern, with minute human intervention. The machine kinematics of both W.C Fields and inclined plane automatons oscillate in a fixed cadence and range of motion. After powering on the motor, an operator only needs patient timing to place each ball one by one into the automaton's claws to initiate the juggling pattern. The illusion of juggling is revealed as clawed hands move in the same way with or without balls.

[20] Beek and Lewbel, "The Science of Juggling."
[21] "The Juggling Edge - Pantograph Juggler."
[22] "Juggling Robot | Hackaday.Io."

Jugglers have also explored 2D planes as performance environments such as in billiard juggling, Kouta Ohashi's Pythagorean Juggling, and angled surface roll juggling. All three roll juggling styles reduce gravity and allow the juggler to perform with a larger number of props. The billiard table walls enclose the balls such that they do not fall and patterns are driven by momentum from bouncing off the walls. Pythagorean Juggling involves an inclined plane with gridded holes such that dowels may be inserted to support various arrangements of bouncing rubber band patterns. The use of marbles requires finer motor movements especially with patterns where the webbed interdigits of fingers are used for catches and throws. Angled surface roll juggling has no table edges so balls are caught underhand as they roll off the bottom edge and tossed onto the table precisely so that they may roll down in a timed arrangement.

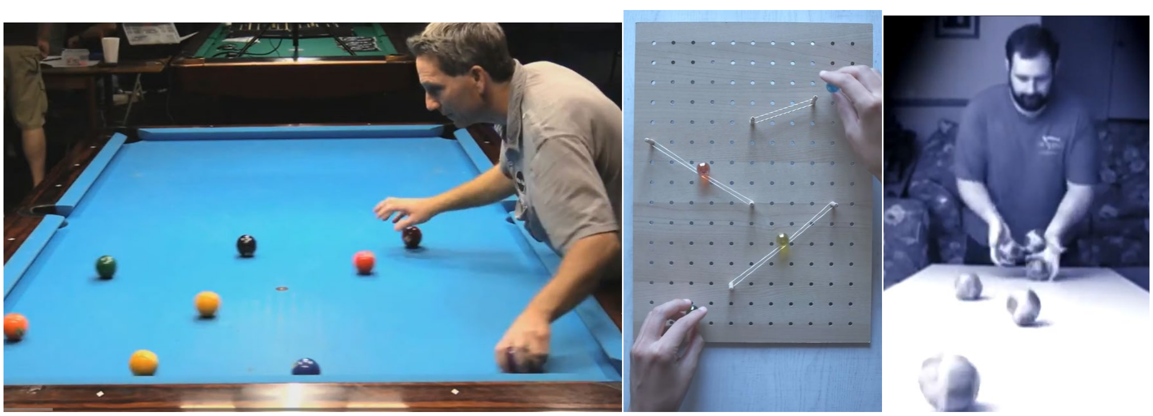

Figure 7. Comparison of surface juggling styles. Left: Tom Kennedy Billiard Juggling, Screen capture by author, "https://www.youtube.com/watch?v=pQyokJBGeyE"; Center: Pythagoras Juggling, Screen capture by author, KoutaOohashi, "https://youtube.com/shorts/f76Q9Vavle0"; Right: Angled surface juggling, Screen capture by author, thegoheads, November 2011, https://www.youtube.com/watch?v=pdgG3zV_2Ak

Feedback in Juggling Automaton Design

In traditional solo juggling there is a feedback communication relation between the juggler and their props. As the juggler learns to perform a new siteswap pattern, their hands must communicate through the language of forces and gravity to interface with balls. Failure to perform a juggling trick results in dropped balls, the main form of feedback for jugglers. Hand-eye relations slowly develop muscle memory to execute patterns without "missed packets".

We can also consider a communication relationship between the Techno-juggler and their automaton during the design and engineering stages. Mirroring the feedback of solo-juggling, there is a similar degree of trial-and-error exchange in which the juggler-technologist may adjust parameters like motor speed, claw size, ball weight, and ramp incline to experimentally test the automaton's ability to perform a desired pattern. This design relation through feedback converges from a shared exchange toward the autonomy of a juggling machine.

Juggling automatons perform well with linear cascade patterns but haven't yet reached a fluency with the variety of siteswap patterns performable by humans. Nathan Peterson expressed the difficulty in enabling an automaton to juggle a siteswap like 534 because each hand is no longer doing the same movement. The mathematical complexity grows as equations for ball positions and velocities become non-linear. While the mathematical languages of equations and spacetime graphs inform a translation between juggling pattern and mechanical automaton design, it didn't reach the level of variety and communicability as did Siteswap Animators.

Siteswap Animators

Simulated computer graphics siteswap animators have had much greater success in representing and adding onto the vast lexicon of juggling tricks. Computer graphics lends itself to more illusion, as animations need not perform under the same physical constraints as material props and mechanical limbs. As siteswap notation became a common means to share juggling patterns in a pre-video internet, siteswap generators forged an early means of exploring and visualizing new juggling patterns. The computer graphical automaton forms a more centralized two-way relationship with jugglers through the common language of siteswap. Serving as a tutor for jugglers to train on new tricks.

Jugglers specifically were among the group of early home computer users and formed online communities like rec.juggling which supplemented existing sharing of juggling news like Juggler's World Magazine which often featured community submitted pieces and news. It was on rec.juggling in the 1990s that juggling computer programmers including Jack Boyce, Allen Knutson, Ed Carstens, Bruce Tiemann, Bengt Magnusson, Joel Hamkins and others built and shared the first siteswap-based juggling animators.

Unlike the mechanics of WC Fields, site swap generators are completely independent of a physical working knowledge of juggling. Hand positions and ball paths may be procedurally generated without need for a direct physical juggling source. Of course, it was due to the work of early techno-jugglers to translate their own working knowledge of juggling into a computer-friendly form. In siteswap programs, the intermediary of the keyboard and program GUI served as a translation mechanism often moving back and forth between the Newtonian language of bodies and gravitational objects and its computer syntax. In fall of 2025, three decades after the emergence of siteswap generators, they went viral on contemporary social media platforms most notably on Tumblr.

On Tumblr's '#Juggling Lab' community, there are hundreds of Juggling Lab outputs which often diverge from the original intent of sharing siteswaps and into the realm of impossibility and jokes. It is through this imagination generated by Juggling Lab users that we may further untangle communication relations between users and virtual bodies at the site of keyboard interfaces.

Chapter 2: Juggling Lab

We can probe further into this relation of users with virtual bodies through an examination of Jack Boyce's software Juggling Lab and its prosumer community on Tumblr. As this digital community's popularity peaked in fall of 2025, the program's original use case of animating siteswaps widened as non-jugglers playfully exploited the interface's boundaries to create memes. Juggling Lab through its diverse means of production, unbounded parameter space, image props, and mathematical syntax stimulate users while establishing a freely associative graphics environment.

Jokes & Esolangs

Before moving forward, it can be helpful to consider Geoff Cox and Alex McLean's description of 'jokes as diagrams of innovative action that represent how humans can diverge from social norms.' [23] To show how jokes permute software development, they introduce esoteric programming languages like brainfuck and bodyfuck as two esolangs which intentionally deviate from syntactical norms of traditional programming languages. While brainfuck is intentionally torturous in its punctuation-based notation system, bodyfuck permutates this by forcing users to type such programs through body gesture recognized computer vision instead of the keyboard.

By altering its means of typing, bodyfuck moves language away from the keyboard and into the body itself. In contrast, Juggling Lab produces animated group and individual bodily gestures through numerical sequences for prop movements, contextual signifiers through image props, and vector defined bodily positioning and rotation. This visual output is determined by a

[23] Cox and McLean, "Not Just for Fun."

numerical notation system for movement patterns that choreographs multiple stick figure bodies, generating significations through image props. In this way it reverses the user interaction in bodyfuck, where users perform body gestures instead of typing to construct programs. Surprisingly, a program created initially to aid in the discovery of new juggling patterns finds itself repurposed into an exploratory medium for virtual body expressions. The Juggling Lab Tumblr community shows how the siteswap and vector-based body choreography language lends itself not only to the realm of humor, but also in the formation of a recursive public; A shared social imaginary where alternatives and modifications are made using the Juggling Lab program and community.

Types of Juggling Lab Jokes on Tumblr

In reviewing the hundreds of Juggling Lab GIFs contributed by the Tumblr community over the period of 2025 - 2026, there are some clear types of jokes. This includes for example a self-referential nature to juggling lab users as a community, and analogies to how their experience with the interface is like assembly lines and even juggling jugglers or 'juggle-ception'. There is even the use of the juggling lab interface itself as an image prop to signify the act of using Juggling lab as a form of juggling.

In depictions of group scenes involving 2 to 20 jugglers, group dynamic social relations are recreated like the game duck-duck-goose, stadium waves, monkey in the middle, scenes of witchcraft, third wheeling, and even representations of trickle-down economics. However, there is also the treatment of bodies as a time varying unit, whereby geometric harmony and chaos can be explored. Drawings of regular 3d shapes like cylinders and boxes are generated in the same spaces as depictions of sea urchins and panic attacks. Especially present in these mutated bodies is a recognition that the juggling man's body is without legs. Users responded by positioning multiple jugglers with pass routines to simulate foot juggling inclusive patterns. The stick figure body is also constructed in nonhuman forms like centipedes and spiders

There is a continuum along which Juggling Lab users can render both outputs representative of normative social dynamics, while also stretching into impossible body formations. For example, solo body animations often include common gestures such as those involved in eating, cleaning oneself, playing instruments, and exercising. These also include symbolic social gestures like air guitar, waving hello, hot potato, fighting, playing with cars, taking a smoke break, and giving someone a rose. The choice of image props, bodies, their locations, and looped movements provide meaning to stick figure skeletons bare of a flesh emotional reaction. Also devoid of the fingers, legs, feet, and toes common to the more detailed computer vision anatomies present in Google's Mediapipe and other pose detection models.

Even without the presence of hands, fingers and physical touch, there is still reference to its existence in how the arm joints grasp hammers, guns, boxing gloves, and in petting animals. These examples are not limited to physical touch as they also portray technological interface mediations like the Wii with its gestural remote, a common Xbox or PS4 gaming controller, use of the computer itself, a slot machine, and phone. Further, there is reference to juggling digital communication software interactions such as the icons for taking a photo, sharing a link, typing a text message, and reposting. Additionally, there are computer based software icons used as juggling props like the Microsoft word work in processes, windows power shells, and even the Tumblr logo itself.

Prosumer Production Methods

It should be acknowledged that most of the productions by the Juggling Lab community were generated through one of 3 methods. Typically, users didn't post their program code along with the gifs, keeping them instead as secret. One conversation illustrates this dynamic as:

> fagenthusiast
>
> SERIOUS HOW DO YOU MAKE THESE THE SOFTWARE IS SO HARD TO UNDERSTAND
>
> lolmatdi1
>
> Original Poster
>
> @fagenthusiast Just mess around with it until you get something funny
>
> fagenthusiast
>
> @lolmatdi1 what interface do you use, the jsm or the html or the simple editor
>
> lolmatdi1
>
> Original Poster
>
> @fagenthusiast I just like type patterns I think might be funny in

While there are some who provide siteswap or program code, there is a general treatment of the program as a sandbox for experimental play. Jack Boyce of course provides instructions for notational syntax rules regarding stepwise body and hand movements, and common ball patterns. Many users get started and mess around enough until they land on a funny way to make a joke. It is most common to use the GUI interface to create jokes since the URL based server does not include image props. The Juggling Markup Language (JML) is more suited for users familiar with coding in HTML or XML syntax languages and can allow for greater programmability or organization.

Programmatically Augmenting Juggling Lab

Due to its inclusion of JML, there is a small subset of Juggling Lab users that use other languages like Python to generate JML scripts themselves. This for example allowed for the rendering of the music videos Bad Apple and Never Gonna Give You Up in Juggling Lab through the downscaling of video resolution where each ball represented a single pixel unit. Youtube user "IParts970" was first to introduce this approach by overlaying 432 figures each juggling 2 balls, one black and one white, in a 24x18 ball aspect ratio. Included in their video description is a dropbox link with over 6000 frames from the original Bad Apple video, the jml scripts, and JML output images. There is no presence of additional program files used to generate, but it is likely impossible that they hand-coded 6000+ files each with 432 jugglers and specification of ball pixels as black or white. Likely they ran a downscaling resolution script to take the original video and represent it as a 24x18 binary array and used this to populate a skeleton JML file per frame. Unlike juggling lab animations, each frame is static and the images had to be stitched together into a single video file. Likely they needed another script to open up and save the jpg image for each JML instance for it to be recombined as a video file. Of course, the Rick Roll Juggling Lab gif, made by Tumblr user Foone, likely followed IParts970s method given that they used the same 24x18 structure, but instead of a binary black and white, they would need to include a much wider range of ball prop colors. Foone doesn't attribute any code resources or methodologies reflecting the general attitude of keeping a magic trick secret.

These two users further augmented Juggling Lab with Foone's JuggleKeyboard and IParts970's osu2jml conversion scripts. The JuggleKeyboard is a set of 5 figures pass-juggling 30 keyboard props including 26 english letters and 4 action parameters like delete, space, back arrow, and up arrow. Foone demonstrates how this program emits keypress events to the pc's and can be used to type ‘Hello World’ into the Tumblr Post input box. The severe difficulty being that users need to coordinate their eyes and cursors with the fast-moving keys in order to type out messages. Additionally, IParts970’s osu2jml script connects the community juggling lab to the community of “OSU!”, a PC rhythm game where players coordinate their mouse clicks and drag across rapidly changing sequences of circles called a beat map. IParts970 does provide a GitHub repository with the java project and demonstrates how the juggling lab figure’s arms successfully land balls onto each beat map target object. These examples illustrate how the sensibilities of juggling are associated with computer interfaces of keyboard, mouse, and rapid visual motion.

Through all these examples, among the jokes, there is a certain sublimation depicting user interaction itself. It is in the presence of communication and play interface controllers like Wii remotes, gaming consoles, phones, as well as software application and control icons. Certain augmentations of Juggling Lab also feature alternative keyboard and mouse interfaces which deviate from the gestural norms of sitting at a computer screen and typing. Could it be suggested siteswap's ability to discover new movement patterns still bears resonance when applied outside of physical juggling to virtual bodies and multimedia props?

Chapter 3: CGI Era

In thinking about the formation of a public around computer or machine mediated juggling, there are a variety of formations between technical objects like computer programs and hardware, communication platforms like social networks, and users equipped with a particular interaction behaviors or norms. The juggling community has formed its own publics prior to computers, yet it was among early adopters to technical communication networks as well as development of juggling computer programs. Considering Goriunova's formation of a public through a particular failure or 'something left uncared for by the governing forces of democracy in technological society'. It is interesting to question what failure does a digital juggling public need to respond to and to what degree it 'exhibit[s] a capacity to transform other fields of societal operation without getting locked in them?'[24] To attempt at an answer, it is useful to look at a historically significant appropriation of juggling within early computer graphics.

Motion Captured Jugglers

Initially founded in 1962 as an image processing hardware development company, Information International Inc. (Triple I) began developing computer graphics hardware and software in 1974 under the direction of pioneers Gary Demos and John Whitney Jr.[25] This Motion Picture Project Group produced early computer generated scenes for feature films including Futureworld, Looker, and Tron as well as logo animations for CBS and ABC[26]. Triple-I and the larger computer graphics scene at this time popularized the demo format to promote their business and new computer graphics techniques. It was at the SIGGRAPH

[24] Goriunova, *Art Platforms and Cultural Production on the Internet*.
[25] Demos, "My Personal History in the Early Explorations of Computer Graphics."
[26] Rivlin, "Digital Art/Paint Systems Survey."

Electronic Theatre in 1981 where Triple-I publicly displayed the first realistic digitized figure, a juggler called Adam Powers.[27]

Adam Powers[28] was generated by extracting the motion of juggler Ken Rosenthal who was dressed in a white leotard with black dots painted onto his bone joints. Bodily position and depth were captured by front facing and overhead cameras for which the video frames could be processed and rotoscoped.[29] The parabolic movements of the balls were simulated using physics and timed in sequence to the bodily gestures.

Within the juggling community, this represented a mark in which juggling skill could be made independent from visual output and physical performance possibilities. This singular demo conveyed a future possibility that any juggling trick, and prop could be simulated by a virtual juggler. Art Durinski said:

> "Tomorrow Adam Powers may invent and perform juggling tricks that no human currently knows. Durinski explained that the creation of Adam Powers is an important step on the road to artificial intelligence. When the day of artificial intelligence dawns, machines will "think" for themselves, and a human may be able to type into a computer such simple instructions as "juggle 12 clubs," while Adam Powers figures out the rest."[30]

Adam Powers is 'hard-coded' meaning all the hand and ball movements are set ahead of time and not as conducive to a bi-directional exchange between juggling and its virtual representation. Nonetheless this demo activated an imaginary towards a programmable virtual

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[27] Wu, "THE REALISM OF ALGORITHMIC HUMAN FIGURES A Study of Selected Examples 1964 to 2001."
[28] *Adam Powers, The Juggler (1981) - First Motion Capture CGI Animation.*
[29] Flueckiger, "Digital Bodies."
[30] Bill, "Adam Powers Heralds Age Of Electronic Juggling."

juggler. One where the positions of balls and hands may be generated algorithmically. This reality was witnessed within the next generation of computer graphics.

This relation between motion capture of juggling performance and its virtual representation is also evident in the contemporary VR landscape. Mitama Sakumaru, a Japanese Techno-juggler, produces within the broader context of VTubing. VTubing is common in livestreaming as one's physical embodiment can be translated in real-time to a digital avatar. Sakumaru outlines the production pipeline in two videos on their YouTube channel which consists of a 2 camera lighthouse-style motion capture system tracking positions of both hands, feet, the face, and waist.[31] Interestingly, Sakumaru doesn't address the ball tracking techniques within their videos, instead focusing on the hardware and software used for the four stages of raw body capture, conversion to bone structure, retargeting to a virtual body model, and placement within a 3d scene with lighting and camera perspective. In one of the comments, they address "In my own experience, for example with VR juggling, I did try separating the motion capture part and the software that handles the physics calculations for the balls for general-purpose use, but I ended up doing the calculations within the same software because the slight delay of a few frames was annoying."[32] Strikingly similar to the production of Adam Powers, this mocap pipeline is centered around capture of the body, leaving patterned object motion to physics simulation. Sakumaru's juggling performances are much more diverse with inclusion of rings, clubs, cigar boxes, and other objects as props. There is also inclusion of juggling-specific visual motion language like ball trails, ring connections, and node connections[33].

[31] *[3D Vtuber Explains] There Are so Many Types of Motion Capture [Markerless, Optical IMU, VICON, F...*
[32] *[Complete Breakdown] A Video That Explains the Entire Process of Motion Capture [Vtuber Explanati...*
[33] *The Vtuber Can Do VR Juggling !! - malaVares - Juggling in Virtual.*

Computers are People Too

Whilst Adam Powers provided a new imagination for the digital juggling public as to how emerging technologies like motion capture and computer graphical scene simulation can visualize humanly impossible juggling maneuvers, juggling also belongs to a larger public of performance art. In the early 80's, a narrative formed around computer intelligence as a machine's ability to compute and even invent the mechanics to replicate bodily performance. For artists and entertainers, this condition evokes a fear of replacement or at the very least rearrangement of human position in artistic labor.

Interestingly, the TV program "Computers are People Too"[34], created through Walt Disney Productions in 1982, illustrates a counter-narrative of human-computer collaboration in the performance arts. It samples heavily from the Adam Powers demo and uses the character arc of its host Elaine Joyce, a professional dancer and entertainer, to convey this perspective transformation from replacement to collaboration. There are also segments showing the emerging intervention of computers into practices like music, dance, film animation, education, and gaming. The program's sponsorship from Atari is pronounced through its inclusion of advertisements for the launch of their Atari 2600 home gaming console and Atari 400/800 personal computer. Triple-I is an appropriate partner for the broadcast due to their CGI production involvement in Disney's upcoming Tron movie which they promoted during the segments and advertisements.

One relevant segment focused on Gideon Ariel's computerized biomechanical analysis[35] techniques for athletes and other performance artists. After Elaine performs a pirouette, her

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[34] *Computers Are People, Too! (Complete SFM Network Broadcast, 1982)* 📺.

[35] Ariel, "Computerized Biomechanical Analysis of Human Performance."

computer co-host T.O.M (Telecommunicative operative memory) records and critiques her preparation angle as off by 90 degrees, making it less spectacular. It suggests that such body capture technology can appreciate the beauty of human movement and even offer value judgements, seeing what human eyes cannot. Gideon Ariel furthers this logic through analysis of athletic movement including golf, sprinting, figure skating, boxing, dance, swimming, basketball, and diving. Gideon uses a pen tool to draw skeletal marks frame by frame on the Cal Arts dancing pair, Donald Byrd and Rebecca Bobele, offering a critique of their movements. Through calculation of each skeleton's center of gravity, they address a flaw in Rebecca's jump take-off. The computer shows that her left knee moves too far forward, and she could lift higher if jumping earlier on the take-off. After witnessing the segment, Elaine Joyce becomes worried that all her years of sweating and straining to learn dance were for nothing as computers have reduced performance art to diagrams and numbers. T.O.M reassures us that computers are powerless without humans, they need the heart and creativity of human collaboration to be useful.

The next segment claims that computers expand possibilities for the mind and eye through the growing video game culture. Visually present are scenes of the youth at public arcades turning steering wheels, shifting joysticks, rolling trackballs, turning dials, and clicking buttons to skillfully control the motions of virtual characters in a variety of games like Pac-Man, Frogger, and Star Fire. Advertisements refer to Atari's home console system as the only place to play Breakout, Asteroids, Space Invaders, Pac-Man, and Missile Command. These shots feature excited families and neighborhood communities gathered around the couch taking turns playing the games. As well as an ongoing Scott paper towels Tron sweepstakes for a family vacation to the Tron movie sets at Walt Disney and Hollywood studios.

The broadcast ends with how computers can create a whole new stage or dimension for performance. A stage where computers and humans meet through creative interaction, anything becomes possible in medicine, Industry, education and entertainment. As a conclusion, Elaine Joyce is transformed into the virtual scene of Adam Powers, dancing energetically in duet with his juggling of 3D solids.

Through this broadcast, there are several elements depictive of an emergent time of human-computer interaction through narratives set by the powerful Walt Disney Film and Atari home gaming industries. While the broadcast ensures an optimism for creative synergy between human bodily creativity and computer analytics, it also indicates an intertextuality between computer graphics in the film and gaming industries through the 'Adventures of Tron' video game available on the Atari 2600 just months before the film release. Triple-I and subsequently Disney to strategically frame complex artistic body movements like juggling and dance as quintessential to the new age of computer graphical media. It is paired with gaming and home computing as the terms of interfacing with this human-computer relationship. Yet, the popular mode of engagement through the gaming industries is quite disciplinary in fixing the body through rigid mechanical movements and its positionality away from the public arcade and into the domestic space of the home's couch and TV.

Google Cloud Pose Detection in the 2026 Winter Games

Interestingly, many of these same elements are visible within Google's AI marketing strategy through its partnerships with athletes in the recent 2026 winter Olympic games. Just as the 'Computers are People Too' broadcast deploys body-tracking technology to critique the biomechanical efficiency of performance athletes, Google strategically marketed its ability to help Olympian's 'find their edge'[36] through body-motion computer vision software. Specifically, they refer to the deployment of Google Cloud Pose Detection AI to "see the physics that no one could see". The Olympics included several "Inside the Action with Google Cloud" segments which demonstrated how skiers and snowboarders including Shaun White, Alex Hall, Maddie Mastro, and Colby Stevenson relied upon AI to push physical boundaries in their respective sports. These segments often included frame-by-frame body overlays of skeletons onto the athletes and specific motion metrics like take-off, landing, and rotational speeds. Google includes further optimism of their technology transcending into other disciplines like "Imagine how similar technology from Google Cloud could help a gymnast, a diver, a sprinter. And beyond sports: medicine, robotics, physical therapy—anywhere human movement matters." And "If it could work in a split second at 1440 degrees of rotation, it can work almost anywhere."[37]

Alongside this narrative of collaboration between athletes and visual AI technology to gain a competitive edge in performance were demonstrations of use cases of Gemini Chat AI in web and mobile phones thematically tailored to the Olympic games. Average Google users can learn the history of athletes or even instructional athletic training regimes as shown in the 'Dear Sydney'[38] advertisement. In this way the advertisements serve as a particular demonstration or instruction for how audiences can engage with AI tools. A comparative mapping of the agents:

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[36] Cloud, "Google Cloud x Team USA."
[37] Google Cloud, *Google Cloud x Team USA - Behind the Tech*.
[38] Google, *Google + Team USA — Dear Sydney*.

(Visual Feedback Computer Technique)

Ariel’s Biomechanical Analysis ⇒ Google Cloud’s Pose Detection AI

(Touch Interface)

Atari’s home gaming system and personal computer ⇒ Google Gemini on Mobile

(Realism Depiction)

CGI in entertainment ⇒ Olympians performing with computer collaborators

These advertisements not only inform spectators in the Olympic games through Gemini web interfaces, but they remain intact as general problem-solving mechanisms.

While there is a vast distance in time between the Computers are People Too broadcast and the 2026 winter Olympics, there is a resemblant inclusion of body-tracking technologies used to enhance athletic performance as well as a bodily participation language for prosumers which shifts from the mechanical movements of arcade games to promptable engagement through AI chat tools.

In both ‘Computers are People Too’ and the 2026 Winter Olympics, there is a clear use of visual body-motion analytic tools to correct or increase physical efficiency for the performer. For the athletes, it serves as an instructional feedback marker for pushing their performance into new heights. Prior to computer mediation, athletes trained for new tricks around any mode of gut feelings or fuzzier sensory experiences often difficult to communicate through language. Judgement of athletic performance creates objectivity through numerical statistics like number of spins, heights, and speed. This is a particular form of space-time diagram which has similar logic to labor performance studies of scientific management. To increase efficient factory worker output in automated production, division of labor forced workers to perform repetitive manual tasks. When scoring is reduced to these objective statistics, it affords the use case of biomechanical tools to pinpoint exact physical changes which can lead to faster times and increased rotations.

Interestingly, this resembles the space-time scientific management studies of Frank and Lillian Gilbreth to improve assembly line worker outputs in the early 1900s. By using a motion picture camera with long exposure to track the motions of a small lightbulb fastened to the hand of say a brick layer, the overall motion could be visualized as Cyclograph and reduced to its most essential minimized movements. Sigfried Giedion describes this as "The light patterns reveal all hesitation or habits interfering with the worker's dexterity and automaticity… these wire curves, their windings, their sinusoidal qualities, show exactly how the action was carried out. They show where the hand faltered and where it performed its task without hesitation. Thus, the workman can be taught which of his gestures are right and which are wrong"[39].

These Cyclographs reveal a feedback loop between performer and its relation to itself in performance through motion-tracking apparatus. These models were a means of making the worker motion minded. Such space-time diagrams were often formed into physical wire models, allowing for an objective artifact with which to scrutinize worker motion inefficiency. Efficiency in this case is specific to a reduction in movement efforts and time to increase productive output. In all these technological apparatuses for motion visualization, the performing body is the central subject of analysis. Giedion alludes to this as a "type of phenomena in which the motion means everything, the object performing it nothing"[40] Interestingly, within this same time-period abstract artists including Paul Klee, Duchamp, and

[39]Giedion, *Mechanization Takes Command*.
[40] Giedion, *Mechanization Takes Command*.

Miro engage with bodily motion-curves as a central source for unknown signs and forms "rooted within us, for movement and the symbols of movement become of one flesh with our being."[41]

Care for a Juggling Public

In returning to Goriunova's condition of failure or lack of care as the basis for the formation of a public, juggling is implicated within the shaping of a particular collaborative perception of the emerging computer graphics industry of the 1980s. The intelligence of such a computer system is rooted in its ability to understand a human through a different set of eyes and provide objective instructional correction. Ariel's computer vision technology is not much different from Google's ongoing efforts to demonstrate AI collaborating at the highest physical degree with Olympic athletes, despite a 40-year distance in "technological progress". The objective logic for such computer systems is also rooted in worker motion studies of scientific management, expressing an influence in the aesthetics of performative action as interweaved in the feedback relations of capitalist production. Accompanying this depiction of collaboration is an entertainment industry which establishes its own interfaces or means of engagement with the next generation of computer technology. Just like Gilbreth's workers, these interfaces restrict the user's bodily motion to a minimum from joysticks, buttons, keyboards, and mice to touchscreen mobile devices. For techno-juggling to address such a failure without becoming implicated in such longstanding technological forces, it should consider a shift from computer vision systems based in tracking of the user, and towards that of the prop. As well, the feedback relation between space-time movement and performance need not correct the user towards optimized trajectories but provide non-judgmental space for experimental play with movement.

[41] Giedion, *Mechanization Takes Command*.

Amiga Juggler and Amateur Publics of Digital Juggling

Commodore released its Amiga 1000 in 1985 which became one of the first popular home computer lines continuing throughout the late 80s. Jimmy Maher describes the Amiga as the first true multimedia PC, fostering early forms of participatory culture and open-source software development among amateur creators through production and sharing of audio-visual and 3D computer generated programs[42]. Among the chief marketing demos used to show and sell the capabilities of the Amiga, was a robot juggling three glass spheres known simply as "the Juggler"[43].

The Juggler animation, created by ex-astronomer and software developer Eric Graham in 1986, not only played on Amiga, but also was developed on Amiga 1000 with only 512 KB of memory[44]. The animation's qualities of realism including reflections, shadows, and color gradients responsive to the movements of a juggler and props contributed to its success as a promotional device for the Amiga platform. The juggler served as a sort of spokesperson for the Amiga brand, appearing in the display windows of popular software and computer electronics stores. Also, the animation was featured briefly in Tom Petty and the Heartbreaker's 1987 music video 'Jammin' Me'. Eric Graham had been experimenting with 3D computer graphics professionally on institutional computers for 20 years prior to his work on Amiga and created his own program in the C-language called Spherical Solid Geometry (SSG) to create the Juggler Animation.

While this demo may seem like an isolated creation, Graham distributed roughly 1000 copies of his SSG tool to others in person and by mail via floppy disk. This culture of sharing

[42] Maher, *The Future Was Here*.
[43] amigang, *Amiga Juggler Demo (1080p, 30fps)*.
[44] "Amiga Juggler Animation."

helped spawn a subculture of ‘render junkies’ who created and distributed among themselves images and animations of often stunning quality. Notably, Red sectors Inc. 's Megademo features a call back to the Juggler specifically in its animation of a spherically orbed figure to move in non-physical ways and juggle 4 balls in a half-shower pattern. Additionally, there are a series of animations created by Eric Schwartz consisting of the dynamics of a male juggler and female Juguette as well as a child Juggler, Jr.

Chapter 4: Digital Props

In Techno-juggling, the lines between physical and digital props are blurred. Media objects like images, videos, 3D assets, geometries, and sounds can be linked together with physical objects through object tracking. By analogy, the object tracked balls serve as a form of cursor which can drag media objects around onscreen. But this design approach biases towards thinking of media objects as rigid bodies. Unlike physical props, each digital object contains a unique set of attributes through which it may be manipulated. This section will focus on YouTube videos as digital props and the types of manipulations to perform with them.

Theorists of Digital Juggling

As we have seen, Techno-jugglers incorporated robotics and computer graphics into their community practice throughout the late 80s and 90s. It wasn't until the availability of camera-based color tracking in the 2000s that Techno-jugglers began to consider juggling props as controllers for audiovisual performance in platforms like MAX/MSP. Arthur Waagenar and Joe Marshall both built interactive systems and wrote theoretical papers to explore this new medium for juggling.

Arthur Waagenar's 2009 thesis "Juggling as a controller of electronic music" looks to juggling as a viable solution to a failure he saw in electronic music performance. Similar to problem of audience visibility that TOPLAP[45] addressed through live coding, Waagenar writes:

> "When a human being is present on stage, behind a laptop, the role of the laptop as we perceive it, changes. We immediately assume… the person is merely guarding, or only

[45] "ManifestoDraft - Toplap."

> slightly adjusting the processes going on inside the laptop, a listener might somehow feel deceived. It looks as if the musician is very lazy, not doing anything at all. The listener is puzzled why the 'musician' is present on stage in the first place, and what he is actually doing there - maybe checking his email?”[46]

Juggling, while complex, is visually rich in its ability to signify motion and events for the participants. Risk and failure in the performance of a trick is communicated transparently to the audience members as they can see props dropping to the floor. Wagenaar considers a variety of methods to gather data from jugglers and their props such as with piezo, light, and bending sensors for computers to track juggling events like throws and catches. Ultimately, he moves forward with prop position via video tracking as it provides a continuous stream of two-dimensional coordinate data for each ball instead of discrete events. He programmed a MAX/MSP patch such that “for each ball, the user can control six different samples: for each line one going up, one going down. Juggling three balls offers the controlling of eighteen samples - already more than enough for interesting music”. Wagenaar explores a wide set of musical expressions through this technology in an online performance on YouTube[47].

Figure 8. Vertical height thresholds as sampler triggers. Figure from Arthur Wagenaar in “Juggling as a Controller of Electronic Music, Conservatory of Amsterdam, 2009, 43, Fig. 15

[46] Wagenaar, *Juggling as a Controller of Electronic Music*.
[47] arthur wagenaar, *Juggling Music Arthur Wagenaar*.

Joe Marshall's 2007 paper "Eye-Balls: Juggling with the Virtual" uses a similar ball tracking approach but focuses on how it can be used to form narratives, produce visual art, and play games. Marshall is concerned with the traditional positionality of the juggler as "'slaved' to their music and visual effects, and must perform exactly in time to them, with reduced scope for improvisation and audience interaction."[48]. Instead of choreographing a juggling routine that is timed with respect to a predetermined music track or video-sequence accompaniment, Marshall's system argues for the juggler as activator of their stage surroundings.

Marshall, from the field of HCI, incorporates iterative design and playtesting with local juggling groups to evaluate the effect of his system's "minigames" including geometric visual effects, narrative storytelling, and balls as game controllers. Colored motion trails are a significant visual effect which allow the ball paths to outline motion trajectories often resembling figure eights, loops, hearts, and flowers. An incorporation of delayed motion trails as an aesthetic component is analogous to the Cyclographs discussed in chapter 3. For narratives, he constructed a storyline by reading a script and triggering onscreen imagery that would complement his juggling routine. For example, he played off of juggling tricks names like pistons and factory by including videos alongside the juggling motions. He also remixed the arcade game space invaders such that jugglers could fire at aliens on screen via their ball throws and control the horizontal movement of the spaceship with their body position.

Juggling Datasets and OpenCV Object Tracking Aesthetics

In the 2010's, there was a growth in the use of Python OpenCV library as a means to track props in pre-recorded juggling videos. Movement away from the real-time systems of

[48] Marshall, "Eye-Balls."

Marshall and Wagenaar reflects its function as applying a "filter" over the juggling routine for performance online. Prior to his work with juggling automatons, Stephen Meschke developed ball position tracking Python scripts which he used to annotate hundreds of juggling siteswaps in a database. Color tracking[49] was effective for 3 ball patterns and supplemented with Optical Flow[50] for siteswaps with higher ball counts. In collaboration with other Techno-jugglers online, namely Andrew Olson, geometric visualizations styles were developed by treating balls as nodes to connect via lines and other shapes.

Figure 9. Stephen Meschke's Juggling Dataset. Left: Siteswap Prediction Model, Screen capture by author, Stephen Meschke, "https://www.youtube.com/watch?v=s1DHupsWpNE, October 2017"; Right: Visualization of Juggling Data Set, Screen capture by author, Stephen Meschke, https://www.youtube.com/watch?v=IUCUPkjG3ZQ, December 2018

In 2017, Stephen Meschke created a juggling dataset with over 150 siteswap patterns which includes each ball's xy positions over thousands of video frames. His goal was to "Develop a model that watches a video of juggling and outputs details like trick name and number of catches."[51] Unlike traditional manual annotation approaches, color tracking scripts allow for the balls in video feeds to be labelled automatically. Yet, the need to perform each siteswap cleanly on a consistent video stage is clearly a large undertaking of juggling labor. He was able to generate a neural network from this data using Keras and Tensorflow which could

49 smeschke, "Juggling/Tracking/Colorspaces_tracking.Py at Master · Smeschke/Juggling."
50 smeschke, "Juggling/Tracking/Optical_flow_tracking.Py at Master · Smeschke/Juggling."
51 "Juggling Data Set."

accurately predict the current juggling pattern in a video feed from a set of 3-ball siteswaps including cascade, 423, columns, and left vs. right-handed two ball shower. While this served as proof of concept, it wasn't capable as a generalized siteswap classifier among the hundreds in the dataset. While the ultimate goal of this computer vision labelling task was not achieved, the principle of ball tracking led to numerous new augmented motion aesthetics.

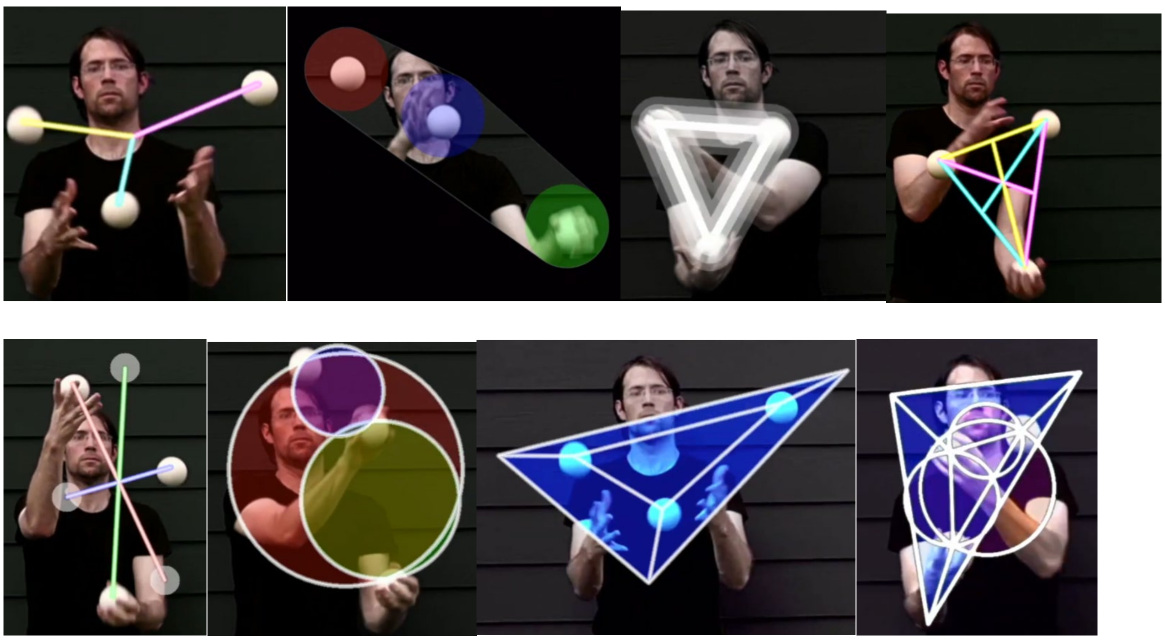


Figure 10. Geometric styles for ball tracking: (a) Of trees, (b) Shallow focus, (c) Triangular Souls, (d) Connectedness, (e) Just A Mirage, (f) The Jewelry Story, (g) Too Sharp, (h) Fragile Stability. Screen captured by author, noslowerdna, "visual effects for juggling" YouTube video, December 7, 2020, https://www.youtube.com/watch?v=CI6Ymmx1fi0

Andrew Olson, known as @noslowerdna on social media, combined his skills in computer science, art, and creative 3 ball juggling to build upon Stephen Meschke's tracking scripts. Alongside siteswap animator experiments in the Juggling Lab community, he regularly produces juggling performances with experimental visual effects. His video "Visual Effects for Juggling"[52] compiles numerous approaches to layering geometric visuals onto a juggling performance. Many of these are node based where lines may connect balls in different ways such as at the midpoint (a) and perimeter (c) which evoke a planar elasticity. Other effects take

[52] noslowerdna, *Visual Effects for Juggling*.

advantage of solid overlapping regions produced like the additive circular colorspaces in (f) and the cropped region in (b). Last is the combining of multiple individual effects as is shown in (h).

BilliArT and DJuggling

There is currently a mixed interest among Techno-jugglers in treating props as sound emitters and nodal sources for motion graphics. One example is BilliArT[53], an audio-visually responsive billiard game produced by Tim Vets and others at Ghent University's multimedia lab. While this installation is specific to billiards, not juggling, it also includes optical tracking of player maneuvered ball props to affect projected visuals and musical expression. In the case of sound, their design incorporates the use of a digital guitar sampler composed of 264 samples for notes strummed up, down, and at two velocity levels. The playback is responsive to the Motion Capturing of each billiard ball's xy position via overhead IR sensing. While the technicality of their Mapping algorithm will not be discussed here, it applies combinatorics to the mapping between ball xy positions and Jazz driven harmonic structures.

The projected table graphics are also visually striking, including a variety of shaped patterns, colored ball trails, vector fields, and gamified targets. In tile shading animations the balls fill the color of geometric diamond and hexagonal tessellation grids as they roll across planar surfaces. Vector field animations show simulated fluid flow field lines based on the distribution of balls. Game targets are also projected onto the table where hitting the ball onto a circle will make it disappear.

[53] Vets et al., "Gamified Music Improvisation with BilliArT."

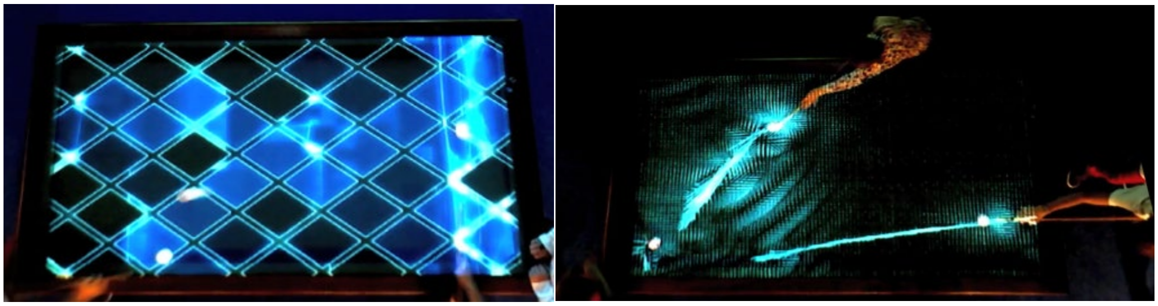


Figure 11. BilliArT Geometric Table Projected Visuals. Left: Tessellated Diamonds, Screen capture by author, Federica Bressan, "https://www.youtube.com/watch?v=U2h_XPKxGQU", October 2015; Right: Fluid Flow Vector Fields, Screen capture by author, Hangaar.net, https://vimeo.com/76552075?fl=pl&fe=ti, 2015

Vojtech Leischner and Pavel Husa's project 'Track me if you can' approaches ball position tracking differently by embedding wireless IMU sensors into the juggling balls themselves. The real-time ball data captured includes airtime, throw or catch event type, rotation Quaternion, throw direction vector, azimuth angle, and altitude. These parameters can be received over OSC or MIDI via a DAW like Ableton which controls sound during live performances. To create music, each ball's sensor can be mapped to a different instrument capable of emitting noteOn and noteOff MIDI messages to a laptop over WIFI. The ball's throws, catches, and airtime are translated to a note's emission and duration. In addition to single notes, pentatonic arpeggios could also be mapped to throws along with "tempo-synced effects such as echo, sync pitch shift, and sidechain compression to enhance the rhythmical aspect of the music."[54] For spatial audio, they can couple a ball's 360 rotation data to panning of the sound sources.

Vojtech has also explored a variety of visual possibilities with respect to digital juggling props. Spacetime braid visualization can plot the balls such that time is the x-axis and ball height is in the y-axis. The plotter accumulates a set history of ball positions, resembling the ladder diagrams mentioned in Chapter 1.

[54] Leischner and Husa, "Sonification of a Juggling Performance Using Spatial Audio."

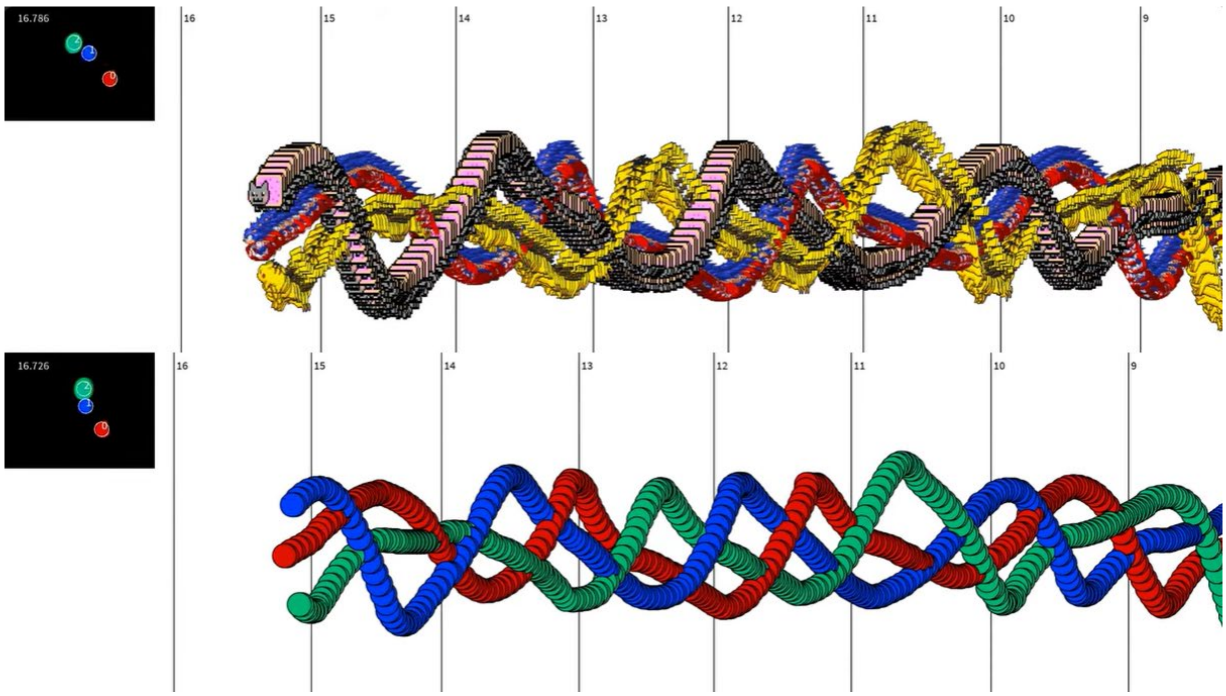


Figure 12. Juggling Spacetime Braids. Screenshot by author from Vojtech Leischner's YouTube video, "Time based audio-movement graph - geometry / sprites juggling," September 27, 2025, https://www.youtube.com/watch?v=tZyrLuXJDk8

Appendices:

1. Iterative Material and Software Design Cycles

Initial design concepts were based on infrastructures of the mobile phone and phone web-apps. The technical library used was Mind-AR[55] which is based on the A-Frame[56] open-source library to build augmented reality applications. Mind-AR can treat images, even non-rectangular ones, as trackable images. When the phone browser's camera detects one of the preset images, it can maintain its tracked position and overlay a variety of other multimedia content typical of augmented reality like 3d models, images, videos, links, as well as audio playback. The YouTube Player API and CSS clip-path values to segment a set of videos onto polygonal film print cutouts. Through this experiment, it was found that the phone's processing capabilities did not keep up with fast changes in the markers, and as marker count increased, it had difficulty maintaining tracking for all. This type of design was intimate as it could include the user's phone to activate interaction, it proved unsuitable the fast movements involved in juggling.

[55] Mindar.Org, "Mindar.Org."
[56] Frame, "A-Frame – Make WebVR."

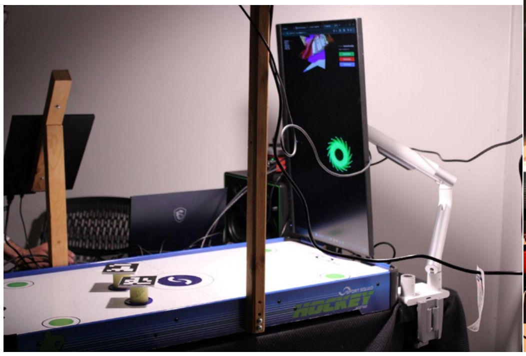
From here the same technical infrastructure was transferred to a computer where an external webcam could be used instead of a phone camera. A gopro camera was tried as well as a cannon DSLR as a webcam. Both camera types require intermediary softwares to enable the camera stream, and gopro is particularly unstable. This experiment revealed that when moving marker images, the main blocker is the occlusion of fingers on top of the tracked markers as they are moved in an overhead camera view. Thus, I tried to use a glass tabletop configuration, with markers face down and camera beneath the glass. This combination worked but comes with the trade-off of needing to illuminate the markers from below but creating disruptive reflection on the glass. Also, this iteration showed that while nonrectangular markers were interesting for user interaction, they significantly reduced real-time tracking ability compared to square Aruco markers. Glass also proved to be too frictional with the markers, making them less fun to move around for the player.

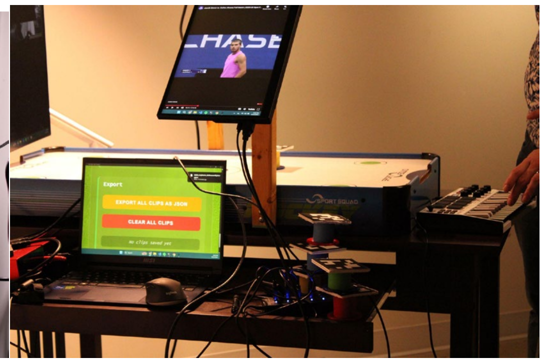
In accounting for the lessons learned in these prior two iterations, it was conceived that the interactive structure needed an overhead camera because of lighting complexities involved with glass, square Aruco markers, and a low friction play surface. The next iteration accomplished this through the modification of a small air hockey table such that Aruco markers were glued on top of airhockey pucks. A vertical mounting arm was made to hold a Logitech Brio 100 webcam in portrait orientation to make most of the air hockey table play area visible. A switch was made to use OpenCV as the computer vision tracking technology and Three.js for augmented materials on an adjacent display monitor. During playtesting with a child, it was revealed that they were not cognisant of the need to make markers visible to the camera and often placed their hands on top of the markers as they threw them across the air hockey surface.

Additionally, it was found that the brio 100 webcam's fps was limited so, I upgraded to an ultra hd webcam supporting 30fps.

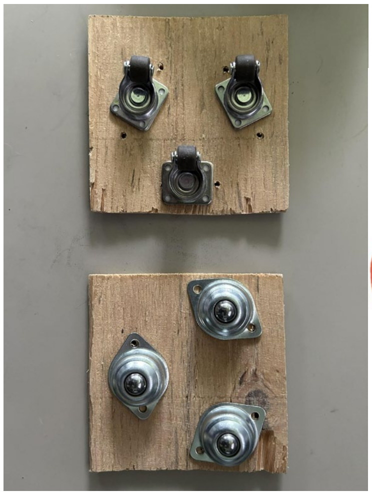
The final physical change was to use cut toilet paper rolls to extend the marker grip area such that the user could hold and throw the markers without occluding the overhead camera view. Also, significant software development was included to support user drawn css clip paths onto youtube videos which could be matched to each onscreen marker. There was a midi controller to move between each video marker to adjust YouTube player-api video values like volume, speed, position, looping, and to transform the videos through RGB color channels, alpha transparency, scaling of size, and index values for collaging multiple videos. The Midi keyboard's piano was used as a sequencer to jump through different sections along the video timeline. Through playtesting feedback, it was found that it was difficult for a single user to play with both the controller and markers simultaneously. Often groups of participants would take on different roles with one throwing the markers around and the other turning MIDI keyboard knobs. This clearly depicts the separation between macro-gestural interaction which could layout spatially the media objects, and common audio-visual control from micro-gestural interaction of musical instruments and keyboard mouse interfaces.

The feedback of this playback experience most significantly showed how user interaction was most prominent when the markers moved against the player. I often took on the role of a second player and passed air hockey puck markers back and forth with the user to bring about more skilled hand eye motions with the user. Also, the time of interaction showed that players needed quick instruction on how to use the midi keyboard. The dials were labelled with their corresponding control parameters, but users often blindly turned knobs discovering how they worked as they played. There was significant difficulty in understanding which video index they controlled as it needed to be set via a drum pad input. Most users didn't have an interest in setting their own videos, as the time needed was more along the 10-15 mark to select videos 2-5 videos, their timestamps, and draw clip-pathed regions. Subsequently, a sampling of JSON example files was made ahead of time and I would choose one for the player upon their arrival.



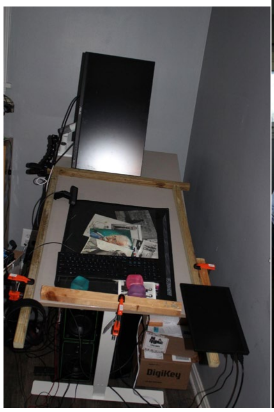
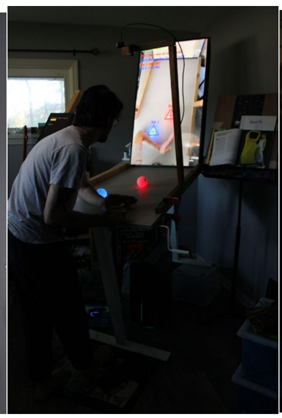
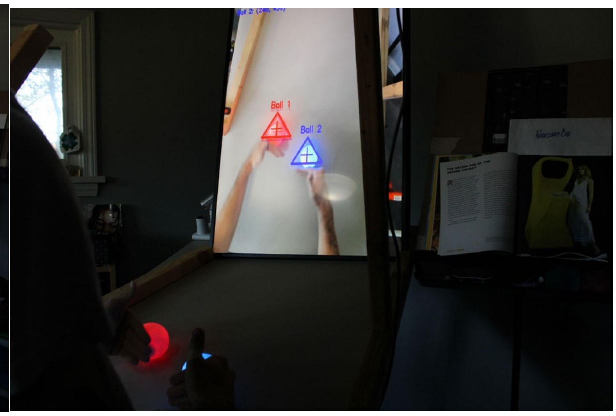


This playtesting experience revealed that a ramped surface would be a most rewarding design change as it created more dynamic parabolic movements and hand-eye coordination challenge for the player. However, to facilitate the types of speed and spatial movements common to juggling, there was also a design tradeoff between increased griping space via longer toilet paper tube and marker-puck friction as the added weight made throwing motions more resistant. At this point it was believed that square Aruco markers most simulated cigar box juggling, which favors rotational tricks, so Aruco markers also worked best to track angle. As it proved too expensive and technically difficult to either get an air hockey table with a higher-pressure fan, 3 new marker-puck designers were tried. The first was square markers made of light wood with 3 ball transfer units screwed to the bottom. This puck was able to roll on the surface and spin, but it was very loud, much heavier, and the bearing friction was still high giving it poor fluid motion. The second design used 3 rotating caster wheels, for which their rotation alignment did not change fast enough to create a fluid experience. The last and most successful marker-puck design was by finding battery powered small fan-integrated pucks which worked by the same principles as the hover-soccer ball toy. These marker-pucks were the most fluid in simulating juggling movements, but when attaching Aruco markers above them, the increased weight still made them resistant and non-erogonomic for juggling grip.

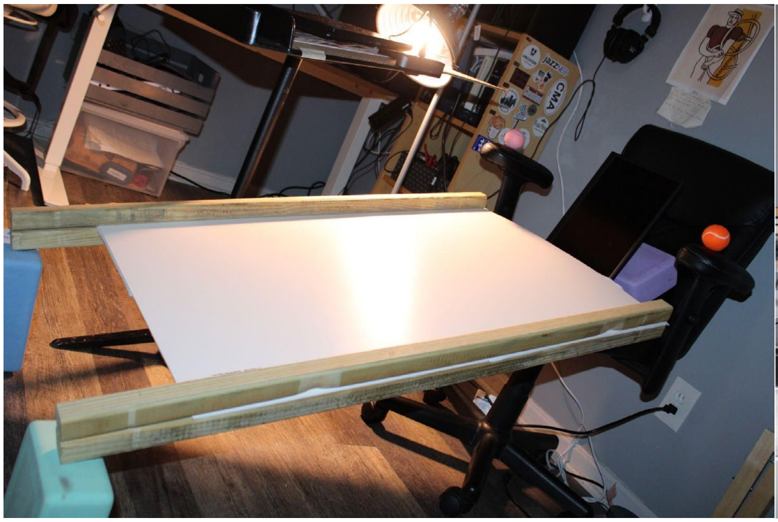



Through these three low-cost experiments, it was determined that commercial wheels, ball bearings, and fan integrated props didn't feel right for table-based juggling interactives with object tracking. From this point, the need for prop-rotation data was abandoned, in favor of interactions which could best simulate the sensations, skill, and diverse possibilities in motion patterns characterized by juggling. Color tracking of a set of color-distinct spheres would be most effective to maintain smooth surface rolling, catching and throwing mechanisms most analogous to the weight and ergonomics of juggling, and potential for color to leak between the fingers, maintaining visibility even during the catch. As a bonus, color tracking is much more computationally efficient than what is required for Aruco markers.

It is simple to switch between an Aruco tracking based Opencv engine and a color tracking one as the software pipeline was already isolated between python backend object position tracking data streamed to the three.js frontend over a Flask Server. At home, I converted my own desk into a design space by tilting one of the legs onto a plastic container such that the desk surface could be inclined. The desk also had an adjustable height mechanism which allowed me to customize it to try out different ramp heights and angles.   I quickly found it necessary to clamp wooden perimeter edges onto the desk to prevent the balls from rolling all around the room during each test trial. Also, I tested different balls including whiffle balls, marbles, contact juggling balls, and tennis balls. I found that tennis balls worked best for their size, weight, and bounciness on the raised table edges.

As it was not a long-term solution to use my desk, it was necessary to replicate it in a lightweight way suitable for travel, quick setup, and adjustability to experiment with a range of player heights and ramp angles. Some two-legged designs were conceived of, but it still seemed hefty to say use two saw-horse legs to hold up the surface. Instead, I had a music stand at home and saw that it bore many of the qualities I was looking for like adjustable height and angle, so I did preliminary experiments by taping cardboard to extend the sheet music holder section and roll whiffle balls along the surface. Next, I constructed a 2x3ft. lightweight rectangular frame from a white pvc sheet and was able to replace it with the sheet music stand using its same mechanical adjustment mechanisms. While it was lightweight, an additional clamp was still needed to prevent the stand from slipping into its lowest setting. Adding the overhead camera arm and adjacent display monitor added more weight to be supported. The stand was supportive, and didn't easily tip, although it could be pushed wrongly as the legs were not very wide.




During this time, the interaction architecture was suitable enough to allow for object tracking of juggling props. More software development time was spent integrating existing computer vision juggling aesthetics like ball connections via lines, rings, and ball trails. A live-coding interface was added such that users could see and control effects parameter code via a keyboard located at the base of the juggling table. A foot mouse was included to try out non-hand-based controls while juggling; however, it was found that this arrangement was quite physical to stand on one foot for long durations to balance while rolling the foot track ball and juggling.

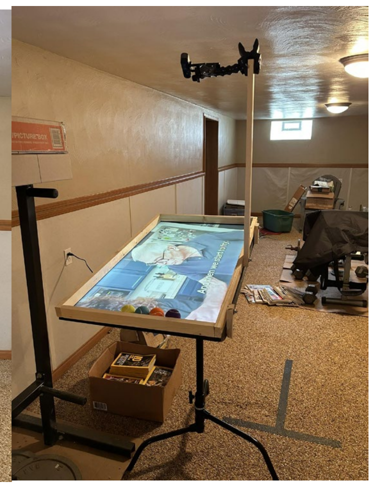
This iteration culminated in the making of a demo video where a variety of these modes were captured along with a growing proficiency in executing four-ball juggling siteswap tricks which I could not perform in traditional upright juggling. While this setup was very good, playtesting with a friend who juggles revealed that a prominent interface impediment was the separation of visual gaze between juggling that happens on the table's surface, and the rendering of graphics on its adjacent monitor. While it was possible to execute more advanced juggling patterns while looking at a camera feeding simulation of the balls, it was more comfortable and skillful to look downwards into the plane of juggling.

There are three choices that exist in achieving such an interaction including Augmented Reality goggles or glasses to produce holographic overlays, projection downwards onto the table surface, and use of a TV as the surface. AR glasses would cost much more significant software development time so it was not considered but may work if others try it in future research. It was easy to try projection by placing a stand mounted projector onto my desk table to give it extra height and face it downwards onto the table. I got good coverage, but felt that it would be clunky, and potentially unstable to have this arrangement. Also, while some work arounds could be achieved, projection inherently could change the color of the balls and risk effectiveness of ball tracking. Lastly, the TV display as a projection surface had most promise as it was learned that most modern TV's bear a polarization layer. This means that circular polarized lenses, oriented with the angle of maximally destructive optical interference, can block out the light emitted from a TV.

This phenomenon was tested using polarized sunglasses with TVs at home, friends' houses, and even in the Best Buy TV section. To execute this trick, one needs only to turn their head sideways while looking at a TV through polarized sunglasses to see the screen gradually darken into blackness. This is not always the case, as not all TVs are linearly polarized, but from a far enough field of view, it will significantly block out most of the content on the display. To move forward with this arrangement, it was decided to invest in commercially available stands and TV mounting adapters. It is not common to mount a TV at an upward angle and in portrait mode, so it was relatively difficult to find the right components for sale online.

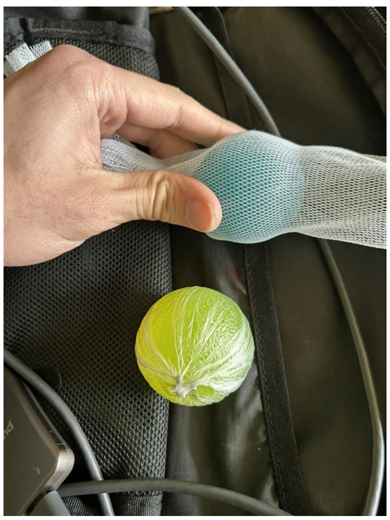
A 43’’ TV was repurposed from another project and the raised perimeter, and camera arm were made modularly attachable to the TV via its feet screw hole sockets. A 58mm circular polarizing camera filter was attached in front of the Logitech webcam using two thin wires wrapped around the lens grooves and twisted at each end for a snug fit. This allowed the lens angle to be freely rotated such that the angle of maximum destructive interference can be set.





With this new arrangement, ball color tracking did decline as the TV’s heavy back lighting darkens the camera’s view. The webcam’s exposure time is already set low at a -5 because it is necessary for high fps tracking performance. Increasing exposure does help color tracking performance, but it becomes much too laggy for real-time play. Two alternative balls were tried with more translucent composition, allowing the ball colors to resonate strongly within camera view due to TV backlight. Colored ping pong balls worked, but their size and weight was too small, inhibiting the ability to juggle. Also, colored squishy balls for carpal tunnel physical therapy were also tried which illuminate well, and have great bounce characteristics, but their rubber composition makes them grip to the tv surface, inhibiting the fluid slide-based throwing motion.

A mini experiment was run to introduce a lower friction layer to the balls including sewing thread, a garlic bag mesh, and plastic grocery store fruit and vegetable bags. It was difficult to tie uniform meridian lines around the ball, which did not roll smoothly. The frictional coefficient of the garlic bag mesh was also too high for it to be rolled smoothly. The vegetable bags worked best, but they always had a small knub on the end melted via heatgun, which created bumpier movements. While there are likely better ways to achieve a smooth layer onto translucent ball props, I could not achieve one through low-cost solutions. In the end, I decided to retain the colored tennis ball solution, but to include an overhead lighting apparatus to ensure adequate top surface illumination of the colored tennis balls.



2. **YouTube Web Interface**

To most, browsing YouTube is an everyday activity on mobile phones, to smart TVs, and laptops in the browser. In its standard web-browser form, the playback of videos is driven dominantly by interface control buttons for volume, speed, captioning, window size, and resolution. Most aren't aware that YouTube also features a set of keyboard shortcuts[57] which add more playback controls. Most notably, one can use the number keys 0-9 to skip to equidistant positions on the playback timeline based on the video's total length. One user, 'nochebynoche'[58] made popular use of this technique for remixing music videos on social media along with drumtracks. Once YouTube has loaded each region of the video, the switching response time is fast enough to allow rapid and rhythmic button presses resemblant of DJ or sonic remix aesthetics. There is also potential for more techniques like speed changing via the "< >" left and right arrow keys, and movement to the next and previous videos in a playlist via the "Shift + N" and "Shift + P" commands. While the keyboard shortcuts provide new modes of interfacing with YouTube videos, they also contain design limits. The number-key division of each video falls into 10 timestamps calculated automatically, with no possibility for triggering the playback of other specified timeline positions. Also, there is limited ability for automatic looping within a specific timestamp region from start to end.

YouTube Player Link Embed Controls

Video embedding is also a popular technique to selectively control YouTube video playback. Embedding allows video player objects to exist on websites outside of YouTube

57 "Keyboard Shortcuts for YouTube - YouTube Help."
58 Is It ?

where playback logic can be programmed through URL parameters[59]. Prior to fall 2025, YouTube embed parameters could be set and viewed directly in the browser via URL search. Recently, the terms of service have changed needing to "to provide an HTTP referrer. If this information is missing, viewers attempting to watch embedded YouTube videos will encounter blocked playback and an error screen ('error 153')."[60] Prior to this change, it was possible to use a URL syntax of "www.youtube.com/embed/{videoId}?start=90&end=95&loop=1". Without the embed keyword, it is currently possible to share a video at a particular start-time but not ending or looping with "https://www.youtube.com/watch?v={videoId}&t=10s". It is also important to note that most copyrighted and monetized videos will not playback in embedded form.

Youtube Player API

The JavaScript YouTube Player API is more flexible for playback controls than is possible through embedded URLs and keyboard shortcuts alone. Its target user base are web-programmers who can integrate the closed-source YT-Player library[61] to arrange videos within their websites or other applications. I used the YT-Player library in early design iterations in conjunction with a MIDI keyboard controller. This allows developers to build out their own audio-visual gestural controls for multiple videos such as analog dials to control volume and speed, while using piano keys to seek timeline position. Also, the integration with web programming lends itself to CSS manipulations like scaling, RGB color shifting, transparency for layering, and clip-path coordinates for non-rectangular video perimeters.

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[59] Google Dev., "YouTube Embedded Players and Player Parameters | YouTube IFrame Player API."
[60] "Embed Videos & Playlists - YouTube Help."
[61] Google Dev., "YouTube Player API Reference for Iframe Embeds | YouTube IFrame Player API."

A relevant web-browser project in this space is "Live Coding YouTube"[62] developed by Sang Won Lee, Jungho Bang, and Georg Essl. Their project is motivated by enabling free improvisation of YouTube source material with respect to the lineage of turntablism, DJ culture, and 60s experimental music. In utilizing the YouTube Player API, they remain critical of the ways in which Google prohibits particular artistic expressions through its terms of service such as API player functionality modification and automated playback detection. Their livecoding interface features an NxM grid based video playback mechanism equipped with video searching by id and name, phase shifting and synchronization, looping, speed-changing, and timeline seeking to name a few.[63] In their paper, they express an interest in "liberating videos from the grid, which will allow more free placements of YouTube videos and realizing advanced visual effects, such as crossfading, cut-out, dissolving, which may enable 'live-coded films'."[64]

Despite their achievements, a key limitation of the YouTube Player API for experimental sound art is the absence of multichannel playback and access. The YT-Player directly routes audio streams from videos to the default PC speaker driver. When multiple video streams are active, they are routed onto the same channel preventing intermediary access to both combined and individualized audio sources. This prevents any possibility for routing digital audio effects such as pitch shifting, delay, reverb, modulation, panning, and frequency filtering and other synthesized audio styles.

[62] "Live Coding Youtube."
[63] nevotgnab, Live Coding YouTube - PAT Showcase 2017 (Screen Recording 2).
[64] Lee et al., "Live Coding YouTube."

YT-DLP

To circumvent these limitations, it is possible to use youtube-dl[65], an open-source community developed library capable of downloading and gaining direct access to the audio and video streams from thousands of web-players including YouTube. It can be installed in its Python form via pip and has an accompanying command line interface. This approach intends that a video source, whether .m3u8 URL, .mp4 from file, or other encoding formats, be used as a source for HTML <video> objects. I commonly used .m3u8 formats through the command "yt-dlp -g -f 18 www.youtube.com/v={videoId}" which returns a direct decrypted video stream from YouTube's servers. Unfortunately, YouTube's encryption scheme is dynamic, so the link needs to be regenerated at hour intervals.

Youtube-dl has faced legal resistance numerous times to try to ban their service through violation of DMCA copyright laws. In 2020, GitHub was forced to take down the youtube-dl repository by the Recording Industry Association of America (RIAA)[66]. In the RIAA's letter, they cite a particular code segment which downloads popular songs owned by Warner, Sony, and Universal Music groups.[67] The Electronic Frontier Foundation (EFF) responded to Github on behalf of youtube-dl, citing that the use of copyrighted content was part of a unit testing script used for version changes, which they subsequently replaced to non-copyrighted videos for compliance. The decryption of a YouTube video's signature is justified as "youtube-dl contains no password, key, or other secret knowledge that is required to access YouTube videos. It simply uses the same mechanism that YouTube presents to each user who views a video."[68] Since then

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[65] yt-dlp, Yt-Dlp/Yt-Dlp.
[66] Harmon, "RIAA Abuses DMCA to Take Down Popular Tool for Downloading Online Videos."
[67] github, "Dmca/2020/10/2020-10-23-RIAA.Md at Master · Github/Dmca."
[68] Electron. Front. Found., "EFF Letter to GitHub on Youtube-Dl Takedown."

youtube-dl has remained accessible and functional through committed community development in a web space with regular breaking changes.

During my own experiments with youtube-dl, my computer was temporarily not detected and IP banned by Google, preventing further use of the service. This was due to generating too many .m3u8 urls in short intervals during code debugging. Also, there is a convenience issue in that it becomes messy to deal with long .m3u8 URLs especially in live-coding environments as they span hundreds of characters and multiple lines. Of course, custom text-editor gestures or features could accommodate this, but these two issues challenged me from continuing to develop with youtube-dl as a media source.

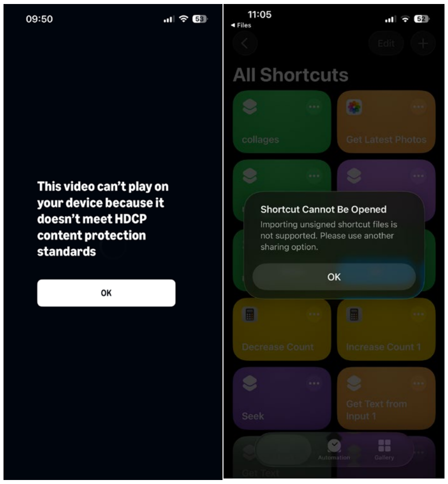
IOS Shortcuts

Considering the opportunities and limitations for performing with YouTube videos through these techniques, an alternative approach is to treat mobile phones themselves as video player sources. Particularly, the iPhone and IOS platform is a notable candidate given its support for Shortcut automations[69]. Shortcuts are programmable scripts introduced with iOS12 which can perform a variety of actions specific to the device and its installed apps. These scripts can be created on iOS devices or macOS laptops using the Shortcut app's block programming IDE. This route is strictly available to Apple ecosystem users, but it provides a unique lens for thinking about how YouTube, Apple, and subsequently its app creators restrict specific creative actions. The main boundaries faced and obstacles overcome in this research include adapter hardware, file security, gestural control, and restricted control blocks.

[69] Apple Support, "Shortcuts User Guide."

Expanding from the unitary platform of YouTube, Shortcuts can potentially target a larger set of popular playback media such as in social media, web, and streaming platforms. Independent of the Shortcuts platform, I first encountered resistance towards mirroring the phone-screen onto external displays like smart TVs. Smart TVs and their peripheral box devices cater towards screen mirroring through Apple's Airplay, Google's Chromecast, and other casting technologies. However, from the hardware side, there are also many adapters which allow users to mirror directly from lightning or USB-C to HDMI[70]. This works great for landscape phone content, but not for typical vertical mode content. Since TV's dominant resting place is in the widescreen orientation, their settings don't support any custom configuration to rotate the content. Instead, vertical content is fitted such that its height matches the TV width leaving much black space on either side. In Techno-juggling, which features a vertically angled TV screen as a juggling surface, it was not feasible to mirror the phone directly in portrait mode. While one could build a hardware-based picture rotation device, it is most practical to use an HDMI to USB-C video capture card[71] connected to a PC to do the orientation transformation. Additionally, most phone applications are fair game for hardware-based screen mirroring with the exception of streaming services which block playback when detecting an HDMI adapter device due to HDCP content protection standards[72].

[70] "Amazon.Com: FEINODI Lightning to HDMI Adapter with USB Camera and Charging Port, SD Card Reader for iPhone, 6 in 1 Camera Adapter to iPhone, Supports MIDI Keyboard, Mouse, HD TV/Projector/Monitor, Plug & Play : Electronics."
[71] "Amazon.Com: Guermok Video Capture Card, 4K USB3.0 HDMI to USB C Capture Card for Streaming, 1080P 60FPS, Compatible with iPad Mac OS Windows, Quest 3, OBS, PS5/4, Switch2/1, Xbox, Camera (Silver) : Electronics."
[72] Wikipedia, "High-bandwidth Digital Content Protection."



The Shortcuts block language IDE is also a limiting factor because unlike higher-level object-oriented programming languages, it requires repetitive manual work to do common tasks like assigning variables, filing data structures, looping through elements, and using conditional logic. Also, worth mentioning is lack of copy-pasting to re-use block sections and tediousness to copy paste and alter URLs in link open blocks. As these programs are .shortcut filetypes, it was considered they could be generated programmatically on a laptop and opened on the phone to run. Others have focused on this effort, most notably Josh Farrant's shortcuts-js project[73]; however, Apple made a file encryption switch with iOS15 which invalidates the opening of "unsigned" shortcut files[74]. Signing is an encryption scheme which can only be performed by an Apple device and prevents the viewing of the backend syntax for signed .shortcut files. Interestingly, there is a workaround to view the .plist or XML-like structure of a .shortcut file by querying this URL address with the shortcut ID:

73 Farrant, Joshfarrant/Shortcuts-Js.
74 fmacchia, "iOS 15 Unsigned Shortcuts."

"https://www.icloud.com/shortcuts/api/records/{record_id}"[75]

The ability to query and view a shortcut's backend syntax takes out all the guess work and is a major steppingstone towards programmatically generating shortcuts. It is still necessary to have access to a mac computer to sign such generated shortcut and can be done with the following command:

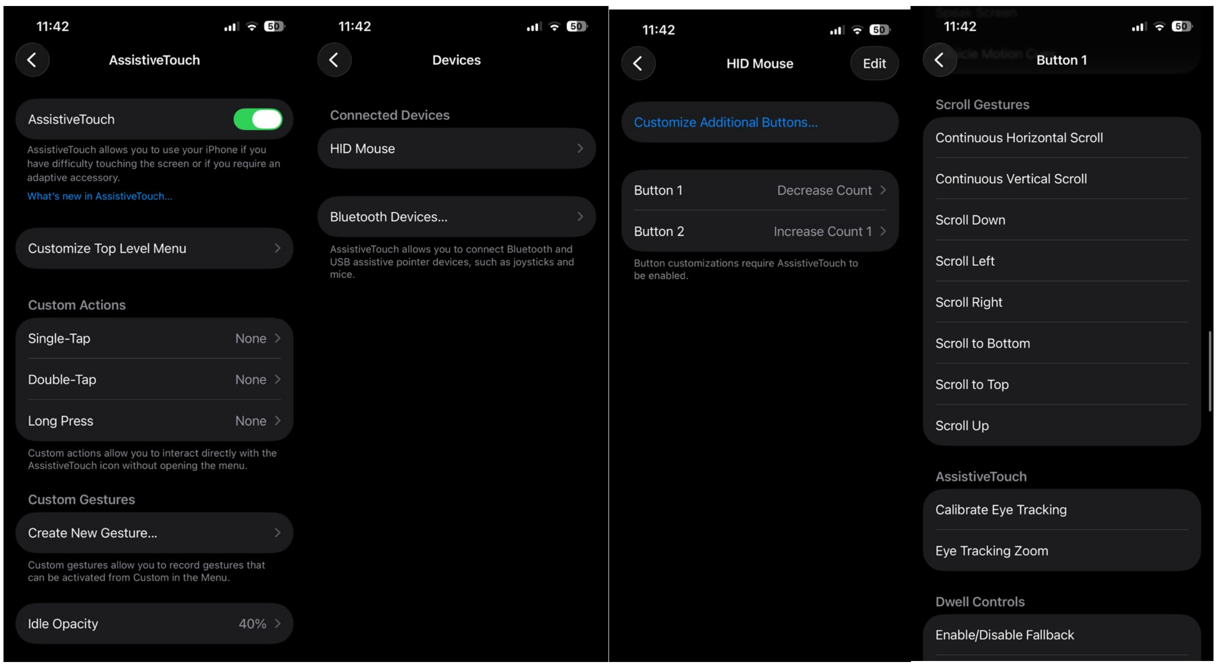
shortcuts sign --mode anyone --input "unsigned.shortcut" --output "signed.shortcut"

This process was followed to generate a shortcut with 20 web and video URL open triggers with custom dwell times for each.[76] Dwell or wait timers serve the role of endtime boundaries and can also trigger loops or more generally next actions.

With regard to gestural controls, Apple and app platform shortcut designers eliminate the possibility for automated touch interactions. For example, it is not possible to open up a web page or YouTube and trigger periodic up or down scrolls, selection taps, or pinch zooming. This decision reflects a broader tendency for popular software to detect and block bot-like behaviors on both web and mobile devices. I could not find any workarounds here but found it interesting to approach this problem by using non-touchscreen USB devices. The prior mentioned lighting port adapter can accept input from common computer mice, keyboards, cameras, and other USB devices. By enabling the phone's AssistiveTouch settings, it is possible to trigger many different device interactions through peripheral button presses such as Scroll gestures and even Shortcut automations. I was able to use an Ablenet BigTrack Footmouse to gesture downward feed scrolling in YouTube and Instagram through button clicks. Also, to trigger index change Shortcut scripts moving "Carousel style" through the previously described programmatically

[75] PythonistaBarista, "Query Shortcut Records API."
[76]Shortcut Juggling.

generated Shortcut. Unfortunately, it was not possible to program the main trackball as it was reserved for cursor position on screen. Further research experiments are needed to assess if Arduino or other microcontrollers can be used to trigger touch interactions on iPhones. This may introduce further methods to transform and automate touch gesture interaction. Also, there is opportunity to scale these experiments with multiple phones.



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